

Introduction to **GWAS**

Data Pre-Processing: Initial & Exploratory Data Analysis

Christian Werner

(Quantitative geneticist and biostatistician) **EiB, CIMMYT**, Texcoco (Mexico)

Filippo Biscarini

(Biostatistician, bioinformatician and quantitative geneticist) **CNR-IBBA**, Milan (Italy)



HerrFaloppio

Oscar González-Recio

(Computational biologist and quantitative geneticist) **INIA-UPM**, Madrid (Spain)



OscarGenomics



Two major genotyping formats (but there are many...)

Plink format ped/map files

<https://www.cog-genomics.org/plink2/>

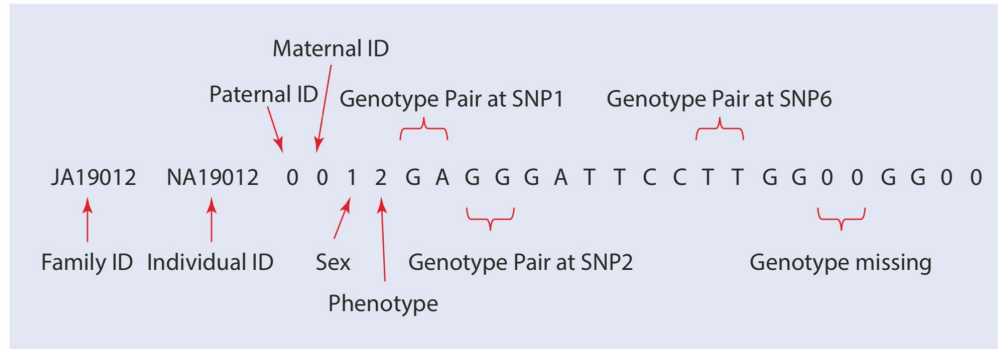
Variant calling format vcf files

<https://www.internationalgenome.org/wiki/Analysis/vcf4.0/>

<https://vcftools.github.io/index.html>



PLINK - the .ped file



Column 1 = Family ID
 Column 2 = Individual ID
 Column 3 = Paternal ID (zero for missing)
 Column 4 = Maternal ID (zero for missing)
 Column 5 = Sex
 Column 6 = Phenotype (1=unaffected, 2=affected, and 0=missing)
 Column 7, 8 = genotype pair of the first SNP1 (zero for missing)
 Column 9, 10 = genotype pair of the second SNP2 (zero for means missing)
 ...
 Column 457393, 457394 = genotype pair of the last SNP228694

Kim J.H. (2019) GWAS Data Analysis. In: Genome Data Analysis. Learning Materials in Biosciences. Springer, Singapore

PLINK - the .map file

		genotype.ped																							
JA19012	NA19012	0	0	1	2	G	A	G	G	G	A	T	T	C	C	T	T	G	G	0	0	G	G	0	0
		1	rs6681049	0	789870																				
		1	rs4074137	0	1016570																				
		1	rs7540009	0	1050098																				
		1	rs1891905	0	1090080																				
		1	rs9729550	0	1125105																				
		1	rs3813196	0	1159244																				
		1	rs6704013	0	1187454																				

genotype.map

- Column 1 = chromosome number
- Column 2 = SNP ID
- Column 3 = Genetic Distance (morgans)
- Column 4 = physical base-pair position (bp)

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Variant calling format - the .vcf file

Data info

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:CQ:DP:HQ 0/0:48:1:51,51 1/0:48:8:51,51 1/1:43:5:...
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:CQ:DP:HQ 0/0:49:3:58,50 0/1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:CQ:DP:HQ 1/2:21:6:23,27 2/1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:CQ:DP:HQ 0/0:54:7:56,60 0/0:48:4:51,51 0/0:61:2
20 1234567 microsat1 CTC G,CTCT 50 PASS NS=3;DP=9;AA=G GT:CQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

CHROM → chromosome/contig

POS → position on chr/contig

ID → SNP name/ID

REF → reference genome allele

ALT → alternative allele (. / 1 or more)

QUAL → Phred scaled quality

-10log10 (call in Alt is wrong)

E.g. 1/10 chance of mistake → 10

FILTER ☐ quality filter (q10 → quality < 10)

INFO ☐ further information

0/1 is not the same as 0|1 !! (unphased / phased)



Some basic data handling – **plink** (run in the shell)

Basic **plink** command structure:

./plink --function specification

Call program from path

Prefix for input files

dogs.ped and dogs.map are the basic input files

`./plink --dog --file dogs --recode vcf --out dogs`

Specify a non-human
chromosome set:
--dog = --chr-set 38

Recode .ped
and .map file
to .vcf file

Prefix for
output files

Some basic data handling – **plink** (run in the shell)

Basic **plink** command structure: ***./plink --function specification***

When ped and map have different names:

```
plink --dog --ped dogs.ped --map dogs.map --recode vcf --out dogs
```

plink reads vcf too!

```
./plink --vcf dogs.vcf --recode --out dogs
```

Some basic data handling – **vcftools** (run in the shell)

Basic command structure: ***./vcftools --function specification***

./vcftools --vcf <path to vcf file> ***--plink*** --out <path to out file>

vcftools --vcf dogs.vcf ***--plink*** --out dogs_plink

(only biallelic markers will be in the output)